

Status

BobaLink Browser Extension — Status

Revision: 15 **Last modified:** 2026-06-13T13:10:00Z **Scope:** BobaLink (extension/) — WXT + TypeScript Manifest-V3 browser extension that detects magnet links and .torrent URLs and forwards them to the Boba merge service on port 7187. **Authority:** master plan docs/browser_extension/IMPLEMENTATION_PLAN.md (9 phases).

Captured-evidence-driven (§11.4.5 / §11.4.45). Every PASS cites a real commit hash and/or a verified test/file artifact. Rows lacking runtime evidence are marked IN-PROGRESS or PENDING — never PASS (§11.4.6 no-guessing).

Baseline facts (verified this session — Session 11)

- **HEAD:** 2011810 (pushed); wave-4 (Phase-9 ci-ext.sh gate + per-store zips, ru locale, content-XSS, credential-leak audit, decrypt-throw parity, tab-group Challenge, and the _locales packaging fix) staged for the next commit.
- **Extension test corpus: 559 Vitest tests across 52 spec files** — same-session npx vitest run → Tests 559 passed (559) — under extension/tests/{unit,perf,stress,chaos,integration,security,e2e} + src/**/*.npx tsc --noEmit clean; npm run lint 0 errors / 0 warnings. (+126 tests over the prior 287/22 baseline: integration 7, security 4 files, stress/chaos 10, locale 4, token suites 8, tab-groups 13, ally 18, group-send nits 2, +more.)
- **WXT build wiring — COMPLETE.** WXT entrypoints exist at extension/src/entrypoints/{background.ts, content.ts, popup/index.html, options/index.html} (thin wrappers over the existing logic modules, so all prior test imports stay valid). npx wxt build produces a **loadable** extension/.output/chrome-mv3/ AND firefox-mv2/:
manifest_version:3, background.service_worker:background.js, popup.html, options.html, content-scripts/content.js, icons 16/32/48/128, AND _locales/{en,ru}/messages.json (REQUIRED because default_locale:"en" + __MSG_*__ — see the §11.4.138 note below). **Every manifest-referenced asset + the default-locale catalog verified present on disk** by the ci-ext.sh manual gate (§11.4.38). Content-script matches are derived from the curated SITE_SELECTORS (24 hosts, **no <all_urls>**); permissions least-privilege (no scripting/tabs/cookies; +tabGroups for Phase-5); host_permissions = http://localhost:7187/* only; CSP script-src 'self'. Per-store zips (bobaLink-1.0.0-{chrome,firefox,sources}.zip) produced by ci-ext.sh.

Session 12 (2026-06-13) — test-coverage breadth + flaky-test hardening + WCAG fixes

Parallel-subagent batch (§11.4.103); the whole batch is verified together by `extension/ci-ext.sh` → **CI-EXT: PASS**, full suite **632 passed (632)** (+73 over the 559 rc baseline), `tsc --noEmit clean`, `npm run lint` 0/0, `chrome+firefox` builds loadable, §11.4.38 `asset-verify` pass, both store zips ≥ 10 KiB.

- **+73 new tests across 4 files** (each anti-bluff, no absolute wall-clock thresholds):
 - `tests/perf/orchestrator-scaling.perf.test.ts` — machine-independent metamorphic sub-quadratic DoS-scaling guard ($t(10 \cdot N)/t(N) < 30$) + golden-good/golden-bad oracle self-validation (§11.4.107(10): `linearRatio=9.9` accept / `quadraticRatio=113` reject).
 - `tests/stress/orchestrator-ratelimiter-tabgroup.stress.test.ts` (7) — orchestrator junk-flood, real `BobaClient` FIFO under a 500-request flood, `TokenBucket` rate-limit engagement, tab-group dedup-batch at scale, + chaos flaky-tab fault injection.
 - `tests/security/infohash-detection-hostile.test.ts` (32) — infohash hex/base32 boundary + case, `xt=` confusion, `.torrent` URL allowlist (traversal/scheme/null-byte), SHA-1 `.torrent` infohash correctness, dedup under hostile repetition.
 - `tests/a11y/focus-and-contrast.a11y.test.ts` (32) — focus management (WCAG 2.4.3/2.4.7/ 2.1.2) + computed colour-contrast (WCAG 1.4.3) from the REAL committed CSS, analyzer self-validated.
- **3 real WCAG AA contrast defects fixed** (`src/popup/styles.css`): `--text-faint dark #7a7a90` → `#838399` (4.07 → 4.61), `light #8a8a9c` → `#6c6c7e` (3.14 → 4.76), `--warning-text light #9a6a00` → `#946400` (4.40 → 4.78). RED → GREEN proven; permanent guards.
- **3 flaky perf/security tests hardened** (§11.4.50 — were FAIL-bluffs under host load, proven load-coupled in isolation, then fixed to assert contention-robust intrinsic statistics): junk-flood security test (tight 5000 ms → 30 s hang-ceiling + scaling moved to perf), `crypto.perf` (p99 → min budget), `parsers.perf` (median → min scaling ratio).
- Full root-cause + evidence for all six fixes: `docs/issues/fixed/BUGFIXES.md` Rev 6 (entries 18–21).

Session 12 — wave-11 (2026-06-13): 6 more coverage files + 5 options-page WCAG fixes

Two further parallel-subagent waves (§11.4.103), verified together by `extension/ci-ext.sh` → **CI-EXT: PASS**, full suite **761 passed (761)** (+129 over wave-10's 632).

Wave-12 (2026-06-13): +38 more tests → **799 passed (799)** (CI-EXT: PASS): `tests/security/crypto-tamper.test.ts` (17 — AES-256-GCM tamper/auth/fixed-IV), `tests/unit/link-scanner-coverage.test.ts` (10 — scheme allowlist/visibility/dedup), `tests/unit/highlight-manager.test.ts` (11 — content-script highlight DOM no-leak/idempotency). No product defects; 1 flaky scaling-ratio test hardened (BUGFIXES 27, §11.4.50/§11.4.118).

Wave-13 (2026-06-13): +15 → **814 passed (814)**. tests/unit/popup-states.test.ts (5) closes a genuine new gap (the popup's **partial Send-All failure** path: succeeded rows flip to Sent, failed rows stay retryable, the live region reports the failure count, Send-All re-enables). Plus **two prior-session test files brought into git** that were sitting UNTRACKED in the working tree (a working-tree-hygiene gap closed): tests/unit/options-save-flow.test.ts (4 — save validation/persistence/encryption) and tests/unit/offline-queue-persistence.test.ts (6 — cross-restart persistence/recovery), both verified real anti-bluff suites against production code. Also: a 6th flaky scaling test hardened (the tab-group **median**→**min** estimator, §11.4.50/§11.4.118 — even a median spikes under 814-test contention; min is the intrinsic-cost estimator). Autonomous test-coverage is now saturated; remaining release work is operator-gated (live-7187 round-trip, headful e2e, store assets).

Wave-14 (2026-06-13) — live-7187 harness prep (de-risks the #1 operator-gated blocker so it turns GREEN the moment the stack is up): the live vitest test now drives the real BobaClient + **independently confirms the synthetic torrent appears in qBittorrent** (:7186 /api/v2/torrents/info) + **cleans it up** (§11.4.14), all SKIP-safe; the live_detect_send_challenge.sh had a real §11.4.14 cleanup-race defect FIXED (BUGFIXES 29); the HelixQA bank audited correct + in sync; and BE-1 (CORS) + BE-2 (.torrent upload) were found **already implemented** in the backend (see RELEASE_READINESS Rev 3). Default suite unchanged at **814 passed (814)** (the live test/challenge are out-of-suite, operator-run when the backend is up).

Wave-15 (2026-06-13) — remaining-blocker prep audits: Blocker #2 (headful e2e) substantially CLEARED — tests/e2e/extension-loads.spec.ts runs + passes **4/4 autonomously** (verified by a real npx playwright test run) via Playwright --headless=new: MV3 SW registers, popup + options render, and (new) the real content script auto-injects on a matched host + detects a magnet + badges it (mutation-proven). **Blocker #3 (store):** STORE_LISTING.md audited submission-ready (all required fields + permission justifications match the manifests; a real locale drift 4→8 FIXED); remaining = operator visual assets + submission. The 6 sibling Challenge scripts audited — all already anti-bluff correct (no changes). See RELEASE_READINESS Rev 4. Default suite still **814 passed (814)**.

- **+129 new tests across 6 files** (each anti-bluff; no absolute wall-clock thresholds):
 - tests/unit/orchestrator-dynamic-content.test.ts (5) — MutationObserver dynamic-content re-scan (insert/href-mutation/relevance-filter/debounce-coalesce/stop-disconnect), RED-proven against mutated production source.
 - tests/a11y/options-contrast-motion.a11y.test.ts (16) — options-page contrast + focus + reduced-motion (found 5 real WCAG defects, now fixed + guarded).
 - tests/i18n/locale-safety.test.ts (41) — i18n catalog safety across all 8 locales (placeholder/key parity, XSS-inert values, JSON integrity); tests/i18n/** wired into vitest.config.ts.
 - tests/chaos/boba-client-resilience.chaos.test.ts (9) — real BobaClient retry/backoff/ timeout/error-classification under fault injection (fake timers, ordering asserts).

- `tests/security/bencode-torrentfile-hostile.test.ts` (52) — bencode parser robustness (truncation, hostile lengths, integer edge forms, deep nesting, wrong-typed info, binary-safe infohash).
- `tests/unit/text-scanner-coverage.test.ts` (6) — bare-text magnet detection, cross-scanner id-equality, false-positive boundary, non-content-node skipping.
- **5 real WCAG AA options-page defects fixed** (`src/options/styles.css` + `src/popup/styles.css`; RED→GREEN, §11.4.120-reconciled guards; BUGFIXES.md 22–26): Save-button gradient 3.66→4.57:1 (button-local), two light-theme-invisible labels tokenized (1.40/1.12 → 10–16:1), reduced-motion blocks added to both stylesheets.
- **1 more flaky perf test hardened** (§11.4.50/§11.4.118): `scanner.perf` + a global `vitest.config.ts` `testTimeout: 30s` so heavy perf/stress tests are never killed by the default 5 s runner timeout under concurrent-suite contention (their real budgets are their own internal assertions). Subagents found **no product defects** in `client/bencode/text-scanner`.

Per-phase status

Phase	Title	Status	Evidence
1	Foundation & scaffolding (WXT config, TS, shared libs, types, constants)	PASS	Scaffolded @33a9815; shared-lib + constants + type-guard unit specs green @15a9a61
2	Core detection & parsing engine (parsers, scanners, infohash dedup)	PASS	Parsers @7225470/fa03323; SHA-1 infohash + link/text scanners + perf/stress @fa03323; scanner orchestrator cross-scanner dedup @946c61e
3	Extension shell — content / background / popup / options	PASS	Shell @e8fde43; background service-worker message router capstone @15a9a61
—	WXT build wiring (entrypoints/, <code>wxt build</code> → <code>.output/</code> , manifest validate)	PASS	Entrypoints <code>src/entrypoints/{background.ts, content.ts, popup/index.html, options/index.html}</code> ; <code>npx wxt build</code> → <code>loadable.output/chrome-mv3/</code> (8/8 manifest assets verified present, §11.4.38); matches derived from <code>SITE_SELECTORS</code> , no <code><all_urls></code> (Session 11 @5edf6ac, pushed)
4	Boba backend integration (real client, queue auth, BE-1/2/3)	IN-PROGRESS	API leaf @e8fde43. Phase-7 decrypt-before-send wired (Session 11): <code>BobaClient.create()</code> decrypts the <code>encryptedBobaApiToken</code> bundle via <code>shared/crypto</code> ; background reads the session passphrase from <code>chrome.storage.session</code> and sends the decrypted plaintext (default-open when locked). Token suites boba–

Phase	Title	Status	Evidence
5	Tab-group batch (chrome.tabGroups → per-tab scan → batch send)	IN- PROGRESS	client-token.test.ts 5 + background-token.test.ts 3 green. PENDING: live-7187 integration (require_backend(7187)), end-to-end detect→send→torrent-in-qBittorrent on the real backend Wired (Session 11): src/tabgroups/ index.ts (dedupe across a group + batch dispatch, 13 tests) integrated into background/index.ts MENU_SEND_GR OUP (deduped group batch → one addMagnets POST); manifest +tabGroups (MINIMAL — research- confirmed tabs NOT needed since only tab.id is read; §11.4.120 security-test reconciliation, mutation-verified). New background.test.ts MENU_SEND_GROUP test (RED- verified: handler no-op → FAIL). Independent review: GO-with-nits. Group-send nits (a) offline-queue enqueue-on-failure, (b) network-error notification, (c) hardened async flush — all FIXED (Session 11, RED→GREEN + 3×-deterministic, independent review GO). TRACKED: decrypt-throw (wrong passphrase) on group send not enqueued/notified — pre-existing parity with Send-All, future phase i18n: locale.test.ts (en completeness) + ru+de+fr+es+it+pt+ja locales added (8 total — plan target reached, 29-key parity, packaged chrome+firefox) with generalized locale-parity.test.ts (key + placeholder parity, mutation-proven); a11y: tests/a11y/ {popup,options}.a11y.test.ts (18 tests — roles/accessible-names/ tablist[?]tabpanel/live-regions, mutation- proven). tablist keyboard-nav (WAI- ARIA Arrow/Home/End). PENDING: deeper WCAG (contrast/focus), theme- switch evidence
6	UI/UX, i18n, accessibility, themes	IN- PROGRESS	crypto.ts adopted; decrypt-and-send path landed + decrypt-throw on wrong passphrase now enqueues+notifies (parity, RED→GREEN); session passphrase from chrome.storage.session, no embedded key, plaintext/passphrase never logged; security suite tests/
7	Security & credentials (delegate-by-default, no embedded key, log redaction)	IN- PROGRESS	

Phase	Title	Status	Evidence
8	Testing to 100% (all types) + Challenges + HelixQA	IN-PROGRESS	security/* (least-privilege manifest, CSP, no-hardcoded-secret, secret-storage, content-XSS — render path proven safe via innerHTML-mutation→FAIL); §11.4.10.A credential-leak audit challenges/security/credential_leak_audit.sh (PASS, mutation-verified); message-router robustness (16 tests) + isValidScanResult scan-result trust-guard + resolveTargetTabId cross-tab confused-deputy guard (fixed a real content-script info-leak: forged payload.tabId could read another tab's detections; sender.tab.id now wins) + scanner-hostile-input (19) + sender-trust (10) + content-XSS. PENDING: full pen-test suite (sender-origin validation, rate-limit) 559 tests/52 files green (unit/perf/stress/chaos/integration/security/e2e). Challenge challenges/extension/detect_and_forward_challenge.sh drives the REAL orchestrator+client end-to-end, PASS on captured evidence, mutation-verified (no-op stub → FAIL). HelixQA boba-bobalink.yaml BOBA-LINK-007 (detect→forward) + BOBA-LINK-008 (tab-group batch) added; symlinked into challenges/helixqa-banks/. tab-group + detect→forward Challenges both mutation-verified. E2E tests/e2e/extension-loads.spec.ts is a real MV3-load test, operator-gated SKIP in this headless sandbox (extension load unsupported; §11.4.3). PENDING: the full 13-type coverage matrix + live-backend integration + the coverage ledger to 100% Manual gate extension/ci-ext.sh (Session 11, §11.4.18 doc'd): tsc → lint → full vitest → chrome+firefox builds → §11.4.38 artifact-verify (opens the manifest, asserts every referenced asset + the default_locale catalog exist non-zero) → per-store wxt zip. CI-EXT: PASS — produces loadable chrome-mv3/ + firefox-mv2/ + bobalink-1.0.0-{chrome,firefox,sources}.zip.
9	Build, packaging & distribution (manual — NO CI/CD)	IN-PROGRESS	

Phase	Title	Status	Evidence
			PENDING: store-listing metadata/ submission + §11.4.65 user/dev/install doc siblings. NO CI/CD (manual only)

Status legend

- **PASS** — implemented and backed by a cited commit and/or verified test/file evidence.
- **IN-PROGRESS** — partially landed; some sub-tasks done with evidence, remainder explicitly enumerated as PENDING.
- **PENDING** — not yet started / no runtime evidence; planned in IMPLEMENTATION_PLAN.md.

Anti-bluff notes (§11.4 / §11.4.6 / §11.4.69)

- The 559 figure IS a same-session recorded `npx vitest run` result (Tests 559 passed (559)), not merely a static `grep` — it supersedes the prior 287/22 `grep-count` baseline. `tsc + lint` captured clean in the same session.
- The §11.4.38 loadable-artifact claim is verified by `ci-ext.sh` **opening the produced .output/chrome-mv3/manifest.json** and confirming every referenced asset AND the `default_locale` catalog exist non-zero on disk — not by “build exited 0”.
- **§11.4.138 bluff-audit (corrected this session).** An earlier Status revision claimed the artifact was loadable on an “8/8 referenced assets present” check. That check was INCOMPLETE: the manifest declares `default_locale:"en" + __MSG_*__`, but the build packaged **no _locales/** (WXT copies static assets from `src/public/`, while the catalog lived at `src/_locales/`) — so Chrome would have REJECTED the extension at load. Caught by the new `ci-ext.sh` §11.4.38 step (which checks the `default_locale → _locales/` invariant). Root cause fixed by moving the catalog to `src/public/_locales/{en,ru}/`; the rebuilt artifact now packages it (verified on disk). Permanent guard = `ci-ext.sh` (the regression can’t recur silently). This is the §11.4.108 lesson: `build-exit-0 ≠ deployable-artifact`.
- The Phase-7 decrypt path is proven by RED→GREEN: ciphertext-on-the-wire (pre-fix) → decrypted-plaintext header (post-fix), with the ciphertext never sent and the token never logged (`background-token.test.ts`, `boba-client-token.test.ts`).
- The detect→forward Challenge was independently re-run and its no-op-stub negation verified (stubbed orchestrator → FAIL) during code review.
- Independent multi-lens subagent code-review (correctness/security/build/docs) was run; the security + correctness + build + docs lenses returned GO / GO-with-nits. (The correctness/security/build subagent lenses were re-run in the conductor context after a transient platform throttle on subagent dispatch; an independent subagent re-pass runs before any release tag per §11.4.40.)
- **Two real bugs caught by the new tests + fixed (wave-5, RED→GREEN, independently re-verified):** (1) a popup²background CONTRACT bug the integration round-trip exposed — the popup read `r.torrent.id` but the real background returns a flat `SendOutcome {id,...}`, so a SUCCESSFUL send threw in the popup and showed a FALSE “Send failed” (row never flipped to

Sent); the popup-unit fake had masked it with the wrong shape. Fixed (popup reads `r.id`; fake corrected; integration assertion reconciled §11.4.120). (2) a `scan-result` TRUST bug the message-router robustness tests exposed — a hostile content script could send `{items:<non-array>}` and overwrite a tab's good detection set with garbage; fixed with an `isValidScanResult` shape-guard (16 robustness tests; the non-array test seeds-good→corrupt and asserts the good set survives).

- de/fr locales added (4 total, all 29-key parity, packaged in both chrome+firefox builds); orchestrator + rate-limiter stress (evidence-captured, deterministic); popup/options[?] background integration round-trips (real `sendMessage`→`onMessage` bridge); USER/INSTALL/DEV guides (§11.4.65 exports).